

Contemporary Technology Development (Industry Case Study Based)

Module 1: Introduction to Contemporary Technology

2-Mark Questions with Answers

Q1. Define contemporary technology.

Contemporary technology refers to the latest tools, systems, and methods developed using modern science and engineering to solve current real-world problems and improve quality of life, productivity, and efficiency.

Q2. Why is contemporary technology important?

Contemporary technology is important because it drives innovation, enhances connectivity, streamlines industries, and addresses modern challenges like climate change, healthcare, and automation.

Q3. Mention two examples of modern technological innovations.

1. Artificial Intelligence (AI)
2. Internet of Things (IoT)

Q4. What is meant by the evolution of technology?

The evolution of technology refers to the gradual development of tools, machines, systems, and processes from primitive methods to advanced digital solutions across different industrial revolutions.

Q5. State two ways technology has impacted society.

1. Improved healthcare and life expectancy.
2. Enhanced global communication and information access.

Q6. What is the role of Artificial Intelligence in healthcare?

AI helps in disease prediction, medical imaging analysis, personalized treatment plans, and virtual health assistants to improve diagnosis and patient care.

Q7. Give one example each of disruptive and sustainable technology.

- **Disruptive Technology:** Smartphones replacing landlines.
- **Sustainable Technology:** Solar panels for clean energy.

Q8. What is the Internet of Things (IoT)?

IoT refers to interconnected devices embedded with sensors and software that collect and exchange data over the internet, enabling smart environments.

Q9. Define the term “digital transformation.”

Digital transformation is the adoption of digital technologies to fundamentally change how organizations operate and deliver value to customers.

Q10. What are two key characteristics of contemporary technology?

1. Integration of automation and intelligence.
2. Real-time data processing and connectivity.

Q11. What is the role of technology in the energy and power sector?

Technology helps improve energy efficiency, integrate renewable sources (like solar/wind), and enable smart grids for reliable power distribution.

Q12. Define Biotechnology.

Biotechnology is the use of biological systems and organisms to develop or make useful products, especially in healthcare, agriculture, and industry.

Q13. What is Industry 4.0?

Industry 4.0 refers to the fourth industrial revolution, which integrates smart technologies like IoT, AI, and robotics into manufacturing systems.

Q14. Mention two applications of Artificial Intelligence in Information Technology.

1. Chatbots for customer service.
2. Predictive analytics for data-driven decisions.

Q15. What is meant by green technology?

Green technology refers to environmentally friendly innovations that conserve natural resources and reduce pollution, such as solar panels and electric vehicles.

Q16. Give two examples of how sustainability influences technology development.

1. Development of biodegradable packaging materials.
2. Innovation in energy-efficient appliances and smart buildings.

Q17. What is a smart grid in the energy sector?

A smart grid is an advanced electrical grid that uses digital communication and automation to detect and react to local changes in electricity usage.

Q18. State one application of AI in the manufacturing industry.

AI is used for predictive maintenance, where machines are monitored to detect faults before they occur, reducing downtime.

Q19. What is personalized medicine in biotechnology?

Personalized medicine uses genetic information and individual health data to customize treatments for each patient.

Q20. Mention two key benefits of Industry 4.0.

1. Increased production efficiency
2. Real-time data-driven decision-making

Q21. What is carbon footprint?

Carbon footprint refers to the total amount of greenhouse gases emitted directly or indirectly by human activities, measured in CO₂ equivalent.

Q22. Name two sustainable energy sources.

1. Solar energy
 2. Wind energy
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5-Mark Questions with Answers

Q1. Explain the need and effectiveness of contemporary technologies in modern society.

Contemporary technologies are needed to address complex global issues such as climate change, urbanization, healthcare demands, and energy consumption. Technologies like renewable energy, AI in diagnostics, and smart transportation improve efficiency, accuracy, and sustainability. Their effectiveness is seen in real-time monitoring systems, reduced operational costs, automation of repetitive tasks, and better resource management. Overall, they help societies transition towards a more connected, intelligent, and resilient future.

Q2. Describe the four stages of technology evolution through the Industrial Revolutions.

1. **First Industrial Revolution (1760–1840):** Introduction of steam engines and mechanized production using water and coal.
2. **Second Industrial Revolution (1870–1914):** Mass production with electricity, assembly lines, and improved communication via telegraphs and telephones.
3. **Third Industrial Revolution (1960s–1990s):** Rise of electronics, computers, and automation in manufacturing and services.
4. **Fourth Industrial Revolution (2000s–present):** Integration of cyber-physical systems, AI, IoT, and robotics transforming industries and daily life.

Q3. Discuss any three modern technological innovations and their applications.

1. **Artificial Intelligence (AI):** Used in smart assistants (like Siri), healthcare diagnostics, and predictive analytics.
 2. **3D Printing:** Applied in prototyping, customized manufacturing, and even medical implants.
 3. **Blockchain:** Ensures secure and transparent digital transactions in finance, supply chain, and governance.
- Each of these innovations has redefined traditional workflows and improved efficiency, accuracy, and trust in digital systems.

Q4. What are the positive and negative impacts of technology on society?

Positive Impacts:

- Improved access to education and information.
- Advances in healthcare, leading to better diagnostics and treatments.
- Economic growth through automation and innovation.

Negative Impacts:

- Job displacement due to automation.
- Increased screen time leading to mental health concerns.
- Privacy and data security issues.

Despite some drawbacks, responsible use of technology can maximize benefits and minimize risks.

Q5. How is contemporary technology shaping the future of work and education?

Contemporary technology is transforming the workplace through remote working tools, AI-driven automation, and digital collaboration platforms. In education, it enables online learning, interactive content, and access to global knowledge. Technologies like Virtual Reality (VR) and Augmented Reality (AR) are enhancing immersive learning experiences. This shift is making learning and work more flexible, personalized, and accessible across geographies.

Q6. Discuss the impact of technology on communication and relationships.

Technology has revolutionized communication through instant messaging, video calls, and social media platforms. It allows global connectivity and real-time interaction. However, it can also lead to reduced face-to-face communication and emotional disconnect. While people stay virtually connected, overreliance on digital mediums can sometimes weaken interpersonal relationships and social skills.

Q7. Explain the role of contemporary technology in addressing environmental challenges.

Contemporary technology helps monitor and reduce environmental impact through:

- **Smart sensors** for air and water quality.
- **Renewable energy** systems like solar and wind.
- **AI** for climate modeling and disaster prediction.
- **Waste management technologies** to optimize recycling.
- **Green building technologies** for energy-efficient structures.

These solutions support sustainability and environmental protection efforts.

Q8. What are the challenges in implementing contemporary technologies in developing countries?

1. **Lack of infrastructure** such as internet access and electricity.
2. **High cost** of advanced technologies.
3. **Shortage of skilled workforce** to operate and maintain technologies.
4. **Resistance to change** due to traditional practices.
5. **Policy and regulatory gaps** limiting innovation and investment.

To overcome these, international collaboration, education, and digital literacy programs are essential.

Q9. Compare and contrast the 3rd and 4th Industrial Revolutions.

Feature	3rd Industrial Revolution	4th Industrial Revolution
Timeline	1960s–1990s	2000s–Present
Focus	Electronics, computers, automation	Cyber-physical systems, AI, IoT
Impact	Automation of manufacturing	Smart, interconnected systems
Example	CNC Machines	Autonomous robots

Both revolutions significantly increased efficiency but the 4th builds on digital connectivity and intelligence.

Q10. How have modern technologies improved quality of life?

Modern technologies have enhanced quality of life by:

- Improving access to **healthcare** and **telemedicine**.
- Enabling **online education** and e-learning platforms.
- Facilitating **remote work** and flexibility.
- Providing **real-time information** for decision-making.
- Making everyday life more convenient with smart homes and devices.

These developments empower individuals and communities with better services and opportunities.

Q11. Explain how technology is transforming the energy and power sector.

Technology enables the use of renewable energy sources like solar, wind, and hydro power. Smart grid systems improve power distribution and minimize losses. Battery storage innovations allow energy backup and stability. Data analytics helps predict demand and optimize supply. Overall, tech adoption supports cleaner, more reliable, and sustainable energy systems.

Q12. Describe the contribution of biotechnology in modern healthcare.

Biotechnology has revolutionized healthcare through:

- Development of **vaccines** (e.g., mRNA COVID-19 vaccines).
- **Genetic engineering** for disease research and treatment.
- **Tissue engineering** and regenerative medicine.
- **Diagnostic tools** like PCR and CRISPR.

These advancements help in early diagnosis, personalized medicine, and improved treatment outcomes.

Q13. What are the main components of Industry 4.0 and how do they benefit manufacturing?

Main components include:

- **Internet of Things (IoT):** Connects machines for real-time data exchange.
- **Artificial Intelligence (AI):** Enables smart decision-making and predictive maintenance.
- **Cyber-Physical Systems:** Integrate computation with physical processes.
- **Big Data Analytics:** Helps in process optimization.

These technologies increase efficiency, reduce downtime, and support mass customization.

Q14. Discuss the role of Information Technology and AI in daily life.

IT and AI are integrated into daily life through:

- **Smartphones** and **cloud storage** for communication and data access.
- **AI assistants** (e.g., Alexa, Siri) for automation.
- **Online banking**, e-commerce, and e-learning platforms.
- **Smart traffic and surveillance systems** in cities.

These tools make life more convenient, productive, and connected.

Q15. How do sustainability concerns drive the development of green technology?

As climate change and resource depletion become major concerns, green technology offers solutions like:

- **Solar panels** to reduce fossil fuel dependence.
- **Electric vehicles** to cut carbon emissions.
- **Green buildings** with energy-saving systems.
- **Waste-to-energy systems** and **water recycling** methods.

These innovations aim to balance environmental conservation with economic development.

Q16. How has AI contributed to smart healthcare systems?

AI has improved smart healthcare by:

- Enabling faster and more accurate diagnostics (e.g., in radiology)
- Predicting disease outbreaks using data analysis
- Managing electronic health records for better patient care
- Supporting virtual health assistants for 24/7 patient interaction
- Assisting in drug discovery and genomics research

These applications enhance accessibility, efficiency, and quality of healthcare services.

Q17. Discuss the role of green technologies in achieving sustainable development.

Green technologies help reduce environmental degradation while promoting economic growth. They include:

- **Renewable energy systems** like solar and wind
- **Eco-friendly transportation** (EVs, hybrid vehicles)
- **Energy-efficient appliances** that lower power use
- **Sustainable agriculture** using precision farming
- **Water purification and recycling** methods

These technologies are essential for reducing emissions, conserving resources, and protecting ecosystems.

Q18. Explain the integration of AI and IoT in smart manufacturing.

AI and IoT work together in smart factories to:

- Collect real-time machine data (IoT sensors)
- Analyze patterns and optimize performance (AI algorithms)
- Enable predictive maintenance
- Automate quality control and inspection
- Reduce human error and operational costs

This results in more flexible, efficient, and intelligent manufacturing processes.

Q19. How does biotechnology support global food security?

Biotechnology contributes by:

- Developing high-yield, drought-resistant crops
- Enhancing nutritional content through genetic engineering
- Reducing dependence on chemical pesticides via pest-resistant crops
- Using biofertilizers and biopesticides to improve soil health
- Enabling faster production of food under controlled environments

This helps ensure a stable food supply despite climate and population challenges.

Q20. In what ways is Information Technology shaping digital governance and public services?

IT enables:

- **E-governance platforms** for public services (tax, utilities, IDs)
- **Online grievance redressal systems** for faster response
- **Digital identity systems** like Aadhaar for welfare schemes
- **Data analytics** for policy planning and implementation
- **Mobile apps** for citizen engagement

These advancements improve transparency, efficiency, and accessibility in governance.